

Supplementary Materials for
Mapping Individual Molecular Connectomes in Alzheimer’s Disease
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Tables S1 and S2

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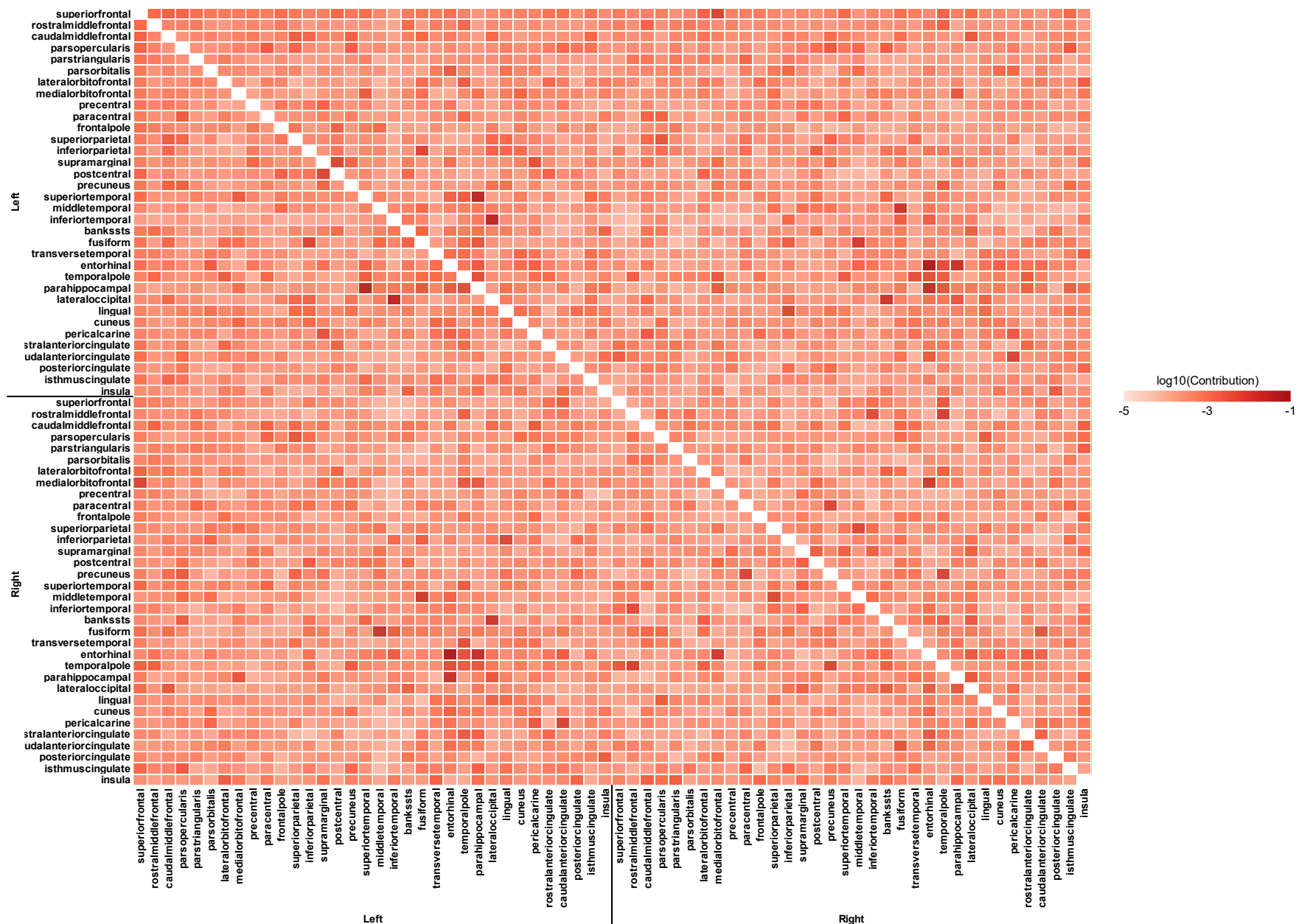
9 **Table S1. Demographic and clinical characteristics of the included participants with tau-**
10 **PET scans after partial volume correction from ADNI.**

	CN Aβ-	CN Aβ+	MCI Aβ+	AD Aβ+	All
N	344	135	143	88	710
Scans	344	239	205	127	915
Age at first scan (years)	70.8 (50.5-94.2)	73.9 (56.5-91.4)	75.5 (56.0-92.4)	76.9 (55.5-93.9)	73.2 (50.5-94.2)
Follow-up scans	-	1 (1-4)	1 (1-3)	1 (1-2)	1 (1-4)
Follow-up years	-	2.1 (0.8-5.4)	2.1 (0.6-4.9)	1.3 (0.5-3.0)	2.0 (0.5-5.4)
Sex (male/female)	150/194	50/85	73/70	51/37	324/386
Education (years)	17.0 (11.0-20.0)	16.0 (12.0-20.0)	16.0 (12.0-20.0)	16.0 (10.0-20.0)	16.0 (10.0-20.0)
MMSE at first scan	29.0 (23.0-30.0)	29.0 (24.0-30.0)	27.0 (19.0-30.0)	23.0 (9.0-30.0)	29.0 (9.0-30.0)

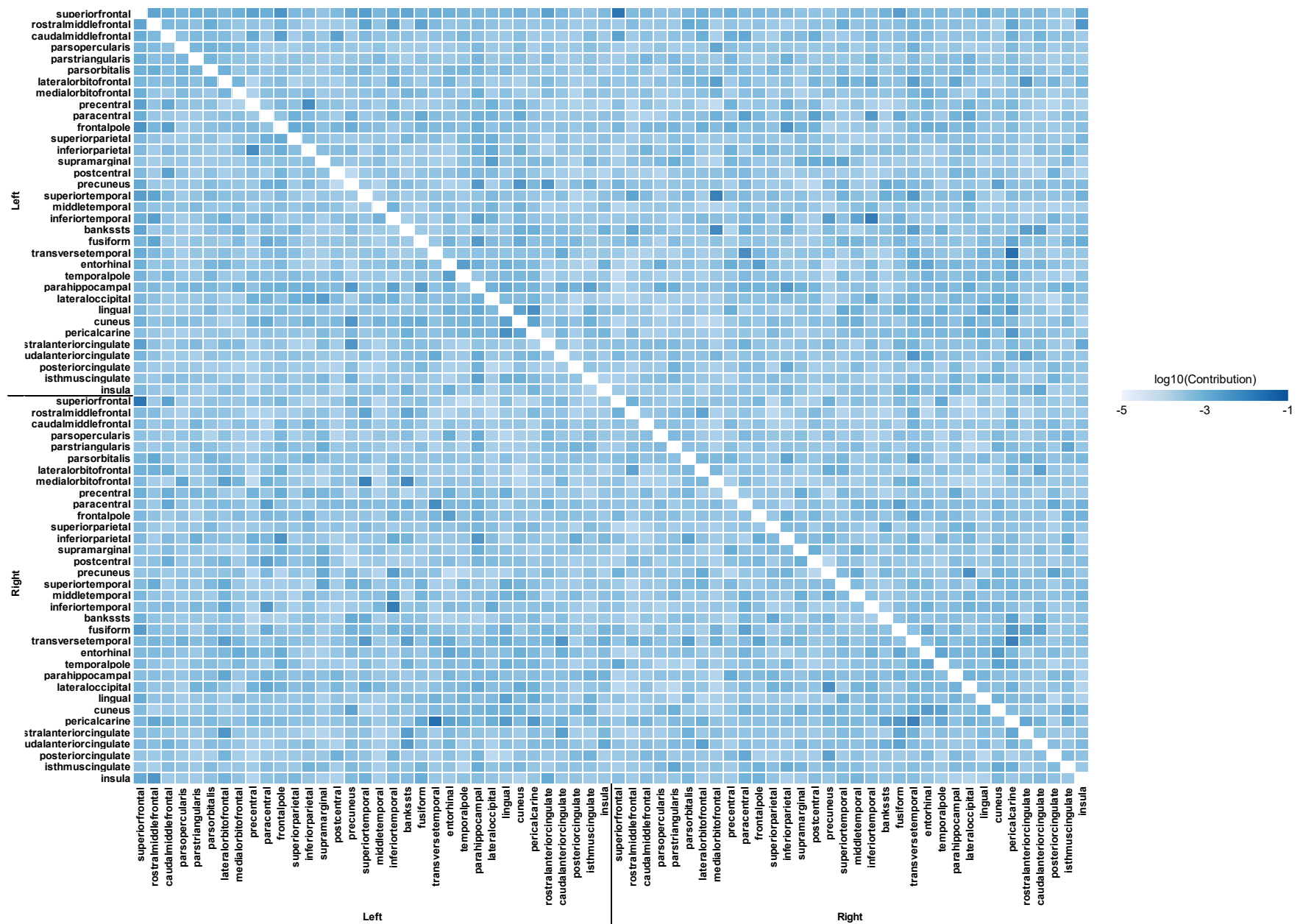
11 Data presented in the table are reported as median (range), unless otherwise stated.

12 **Table S2. Composite brain region label provided by previous studies.**

Brain region	Braak stage label	Data driven tau stage label	¹⁸F-florbetapir global composite region label	Data driven amyloid stage label
superiorfrontal	V	IV	1	II
rostralmiddlefrontal	V	IV	1	II
caudalmiddlefrontal	V	III	1	II
parsopercularis	V	IV	1	III
parstriangularis	V	IV	1	II
parsorbitalis	V	IV	1	III
lateralorbitofrontal	V	IV	1	II
medialorbitofrontal	V	IV	1	II
precentral	VI	V	1	III
paracentral	VI	V	1	III
frontalpole	-	IV	1	II
superiorparietal	V	III	1	III
inferiorparietal	V	III	1	III
supramarginal	V	III	1	II
postcentral	VI	V	1	III
precuneus	V	III	1	II
superiortemporal	V	II	1	II
middletemporal	IV	II	1	II
inferiortemporal	IV	II	1	I
bankssts	-	II	1	II
fusiform	III	II	1	I
transversetemporal	-	V	1	III
entorhinal	I-II	I	1	III
temporalpole	-	II	1	II
parahippocampal	III	I	1	III
lateraloccipital	V	III	1	II
lingual	V	V	1	IV
cuneus	VI	V	1	III
pericalcarine	VI	V	1	III
rostralanteriorcingulate	V	III	1	I
caudalanteriorcingulate	V	IV	1	I
posteriorcingulate	V	IV	1	II
isthmuscingulate	-	III	-	II
insula	V	IV	-	II



13 Fig. S1. Connection's contribution to the discrimination of three diagnostic groups using individual tau connectome.



14 Fig. S2. Connection's contribution to the discrimination of three diagnostic groups using individual amyloid connectome.

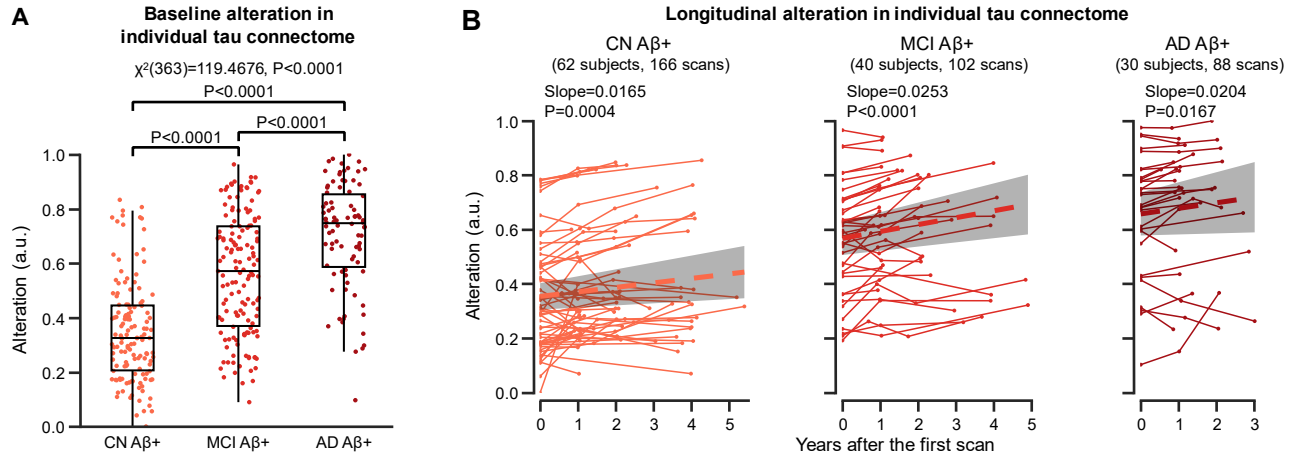


Fig. S3. Alterations in individual tau connectome increase across the AD continuum and over time. (A), The extent of alteration in individual tau connectome across the AD continuum using data after partial volume correction. Each dot represents one subject. Box plots depict the interquartile range and the median value of the distribution. Whiskers extend to the nearest data points ± 1.5 -times the interquartile range. The effects of diagnostic group (CN A β +, MCI A β +, and AD A β +) on individual tau connectome were examined by Kruskal-Wallis test followed by post hoc Wilcoxon rank-sum tests. The significance levels of the Kruskal-Wallis test and post hoc Wilcoxon rank-sum tests were evaluated through 10,000 permutations and were corrected for FDR. (B), Longitudinal alteration in individual tau connectome across the AD continuum using data after partial volume correction. Each dot represents one tau-PET scan. Each thin line connects tau-PET scans from one subject. Bold dashed line and grey shading represent the fitted line and its 95% confidence intervals from the linear mixed-effects model. P values were extracted from the linear mixed-effects model and were corrected for FDR. CN, cognitively normal; MCI, mild cognitive impairment; AD, Alzheimer's disease; A β +, amyloid- β positive; a.u. arbitrary unit.

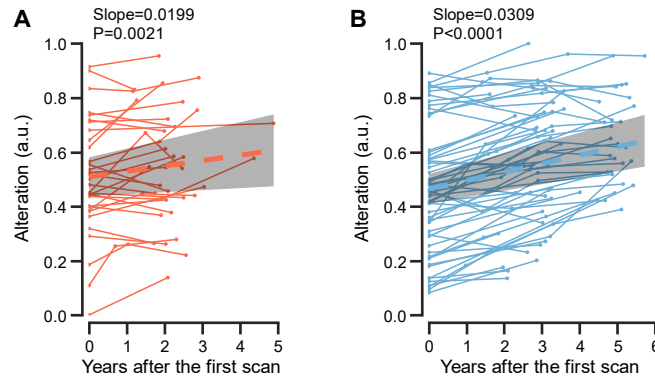


Fig. S4. Alterations in individual molecular connectomes from the HABS cohort increase over time. Longitudinal alterations in individual tau (A) and amyloid (B) connectomes for cognitively normal amyloid- β positive subjects from the HABS cohort (71 tau-PET scans from 32 subjects, 163 amyloid-PET scans from 62 subjects). Each dot represents one PET scan. Each thin line connects PET scans from one subject. Bold dashed line and grey shading represent the fitted line and its 95% confidence intervals from the linear mixed-effects model. P values were extracted from the linear mixed-effects model and were corrected for FDR. a.u. arbitrary unit.

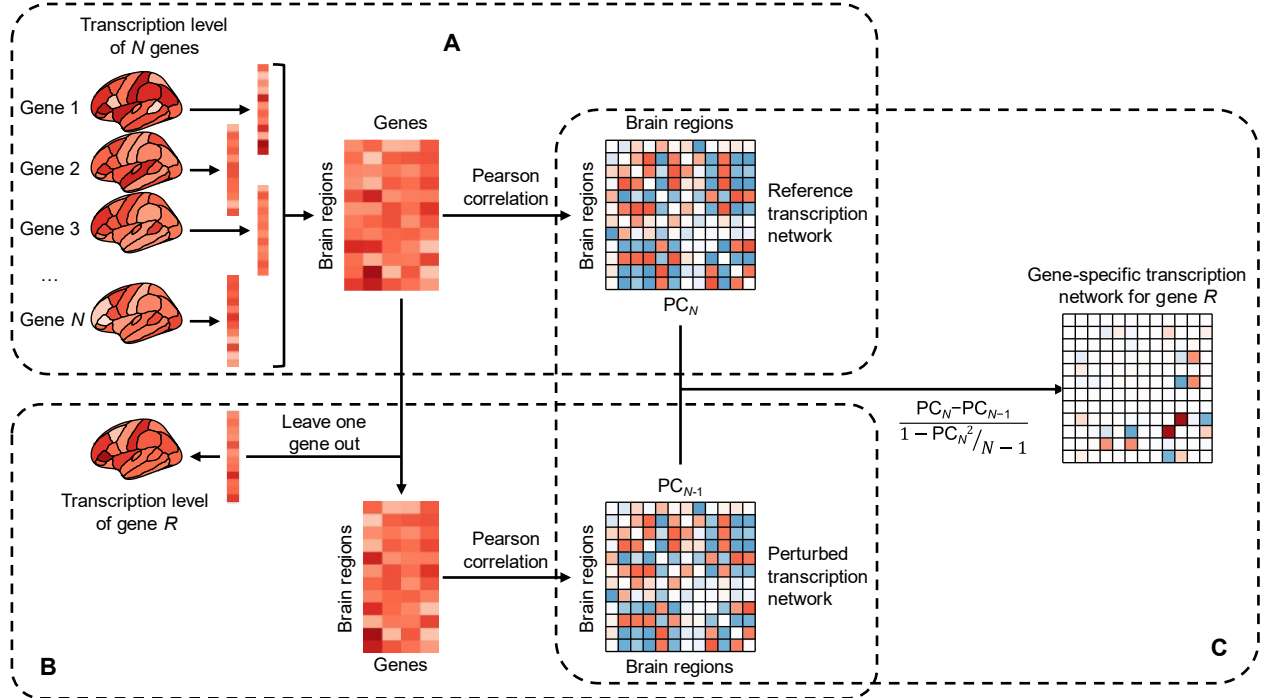


Fig. S5. The workflow for constructing gene-specific transcription network. (A), A reference transcription network was first constructed across $N=10,027$ genes by computing the Pearson correlation coefficient PC_N between transcription levels for each pair of cortical regions. (B), Then a perturbed transcription network PC_{N-1} was constructed by leaving one gene out from the above N genes and re-calculating the Pearson correlation coefficient across the remaining $N-1$ genes. c, The gene-specific transcription network for the left-out gene was obtained through dividing the difference between the reference transcription network PC_N and the perturbed transcription network PC_{N-1} by its standard deviation $\frac{1-PC_N^2}{N-1}$.

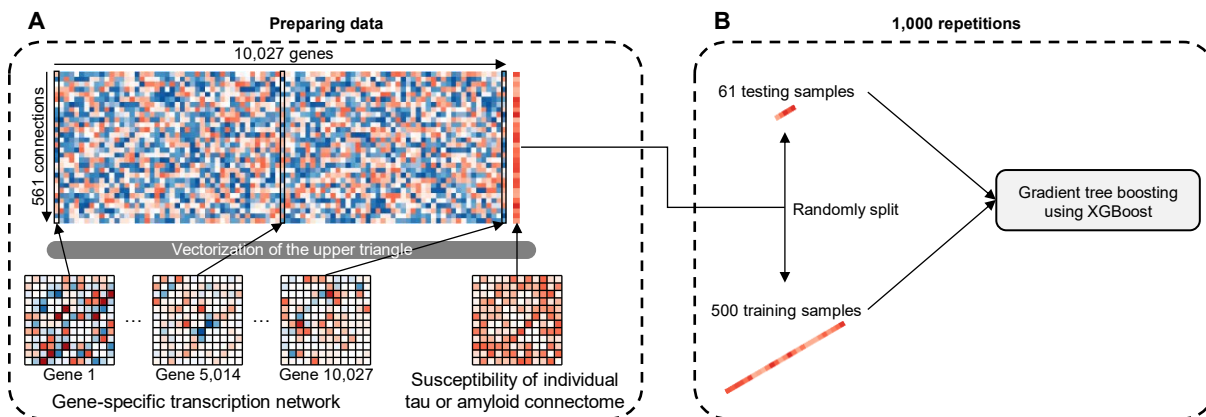


Fig. S6. The workflow for predicting the susceptibilities of individual molecular connectomes to AD using gene-specific transcription networks. (A), The upper triangle of each gene-specific transcription network was first vectorized and was collected into a $561 \times 10,027$ matrix that represents the 561 connections' strength of the 10,027 genes' transcription network. The upper triangle of the susceptibility of individual tau or amyloid connectome was vectorized and was matched with each of the 561 connections. (B), Then these 561 connections were randomly split into training and testing samples that were fed into a gradient tree boosting model to predict their corresponding susceptibilities using the connection strength of the 10,027 genes' transcription network. The randomly splitting training and testing samples and gradient tree boosting model training and testing procedures were repeated 1,000 times.

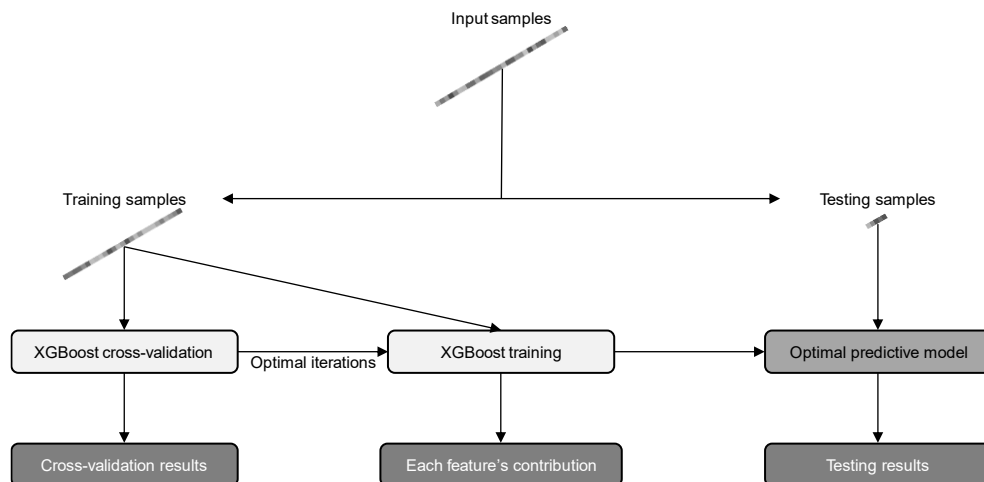


Fig. S7. Gradient tree boosting using XGBoost. The training samples are first used to perform a 10-fold cross-validation training to determine the number of “optimal iterations”. Then, the number of “optimal iterations” and the training samples are fed into the training procedure to obtain the optimal model and the contribution of each feature to the optimal model. Finally, this optimal model is tested using the testing samples.